

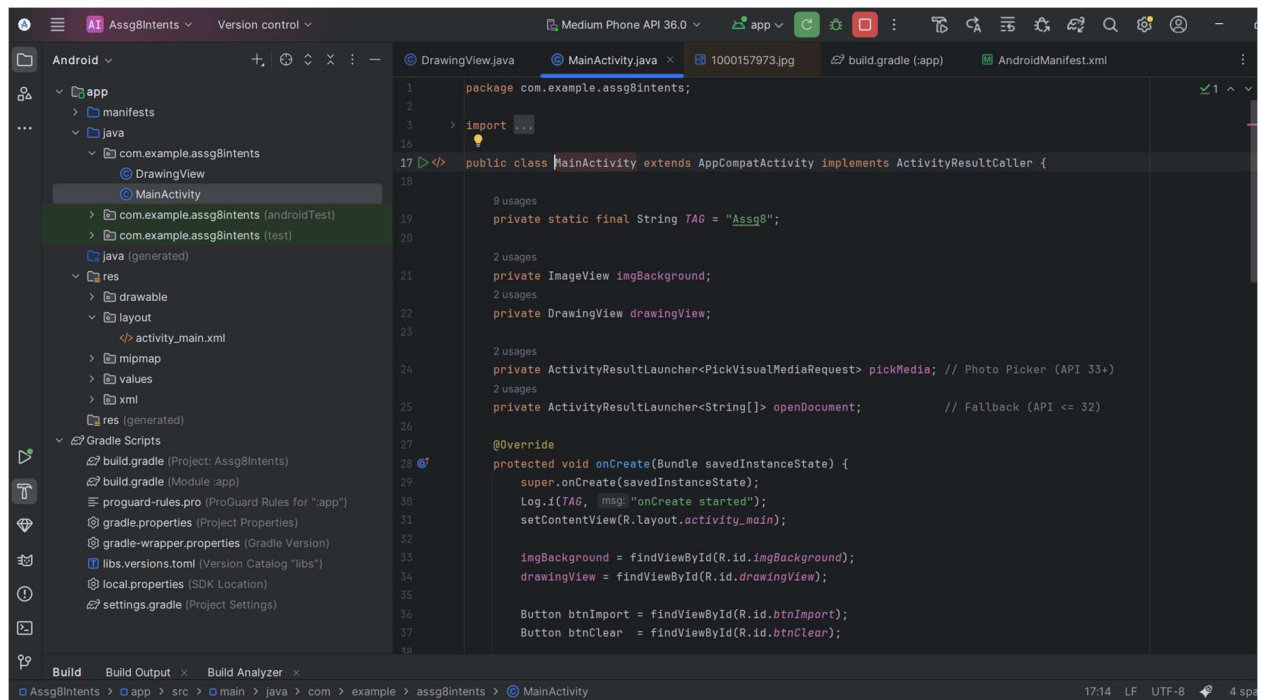
A8 - Intents

Esraa Abdelmoein

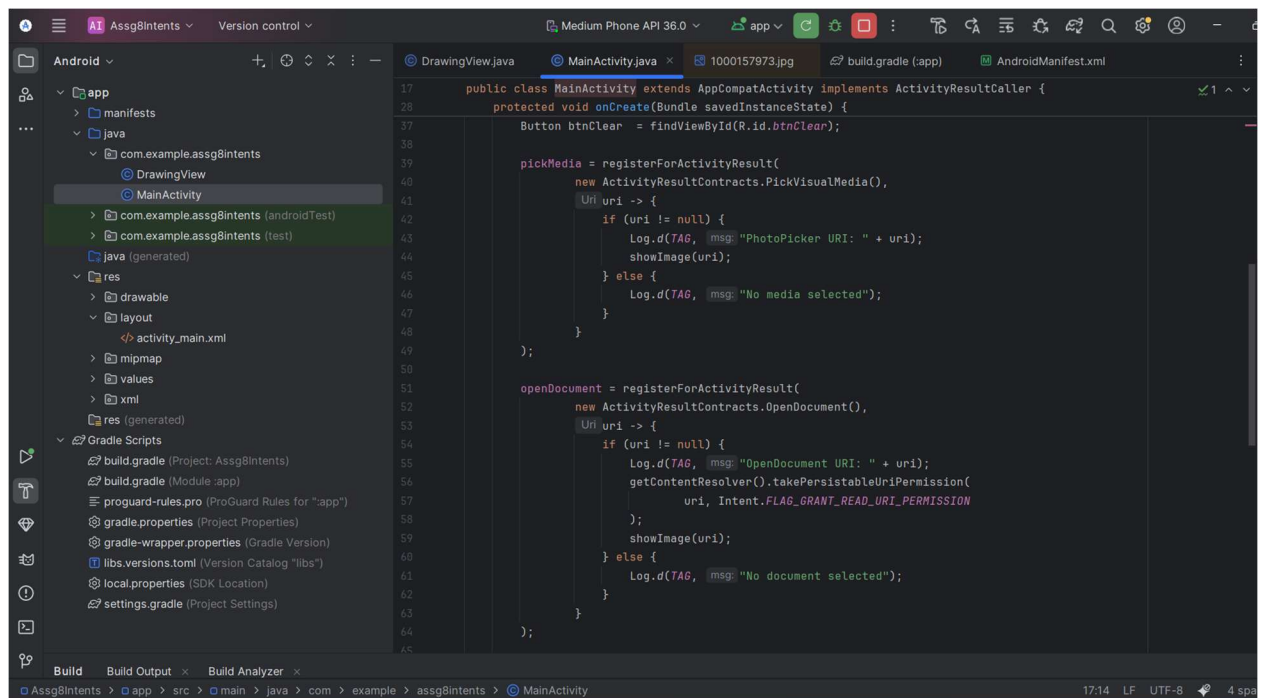
2025-10-29

My GitHub URL : [Mobile-Application-Android/Assignment 8 - Intents at main · EsraaAbdelmoein/Mobile-Application-Android](https://github.com/EsraaAbdelmoein/Mobile-Application-Android/Assignment%208%20Intents)

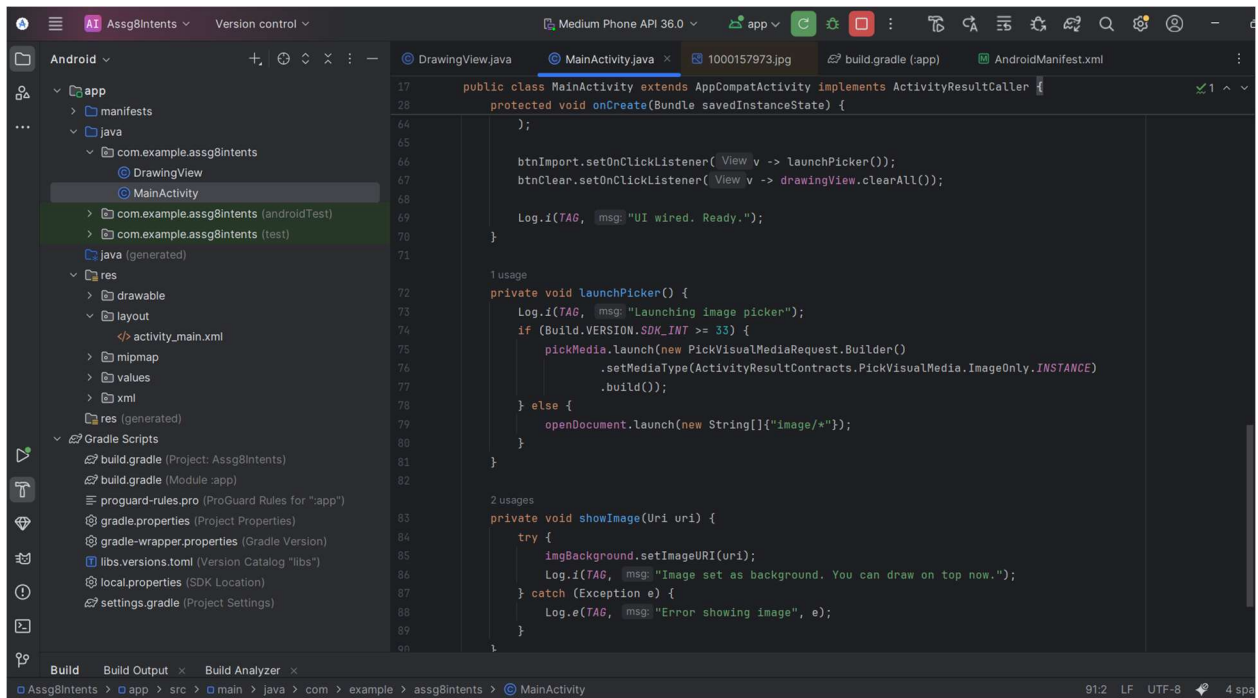
MainActivity.java:



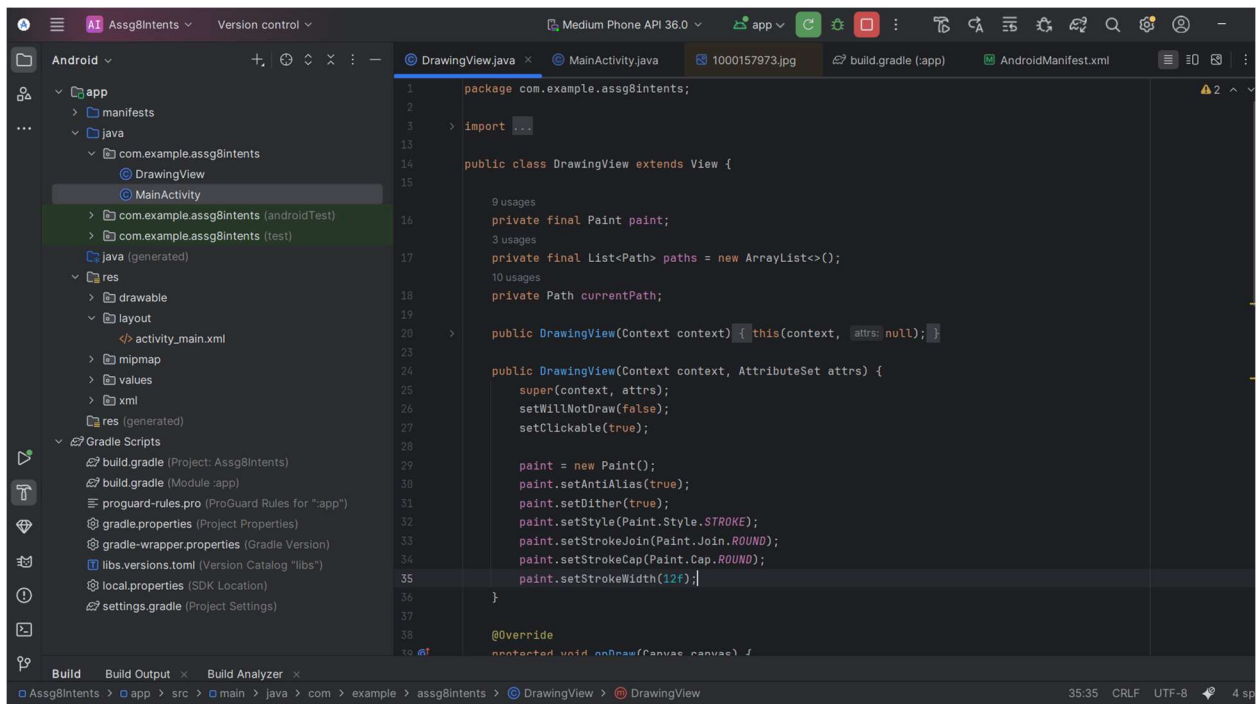
```
1 package com.example.assg8Intents;
2
3 > import ...
16
17 <> public class MainActivity extends AppCompatActivity implements ActivityResultCaller {
18
19     9 usages
20     private static final String TAG = "Assg8";
21
22     2 usages
23     private ImageView imgBackground;
24     2 usages
25     private DrawingView drawingView;
26
27     2 usages
28     private ActivityResultLauncher<PickVisualMediaRequest> pickMedia; // Photo Picker (API 33+)
29     2 usages
30     private ActivityResultLauncher<String[]> openDocument; // Fallback (API <= 32)
31
32     @Override
33     protected void onCreate(Bundle savedInstanceState) {
34         super.onCreate(savedInstanceState);
35         Log.i(TAG, "msg: onCreate started");
36         setContentView(R.layout.activity_main);
37
38         imgBackground = findViewById(R.id.imgBackground);
39         drawingView = findViewById(R.id.drawingView);
40
41         Button btnImport = findViewById(R.id.btnImport);
42         Button btnClear = findViewById(R.id.btnClear);
43     }
44 }
```

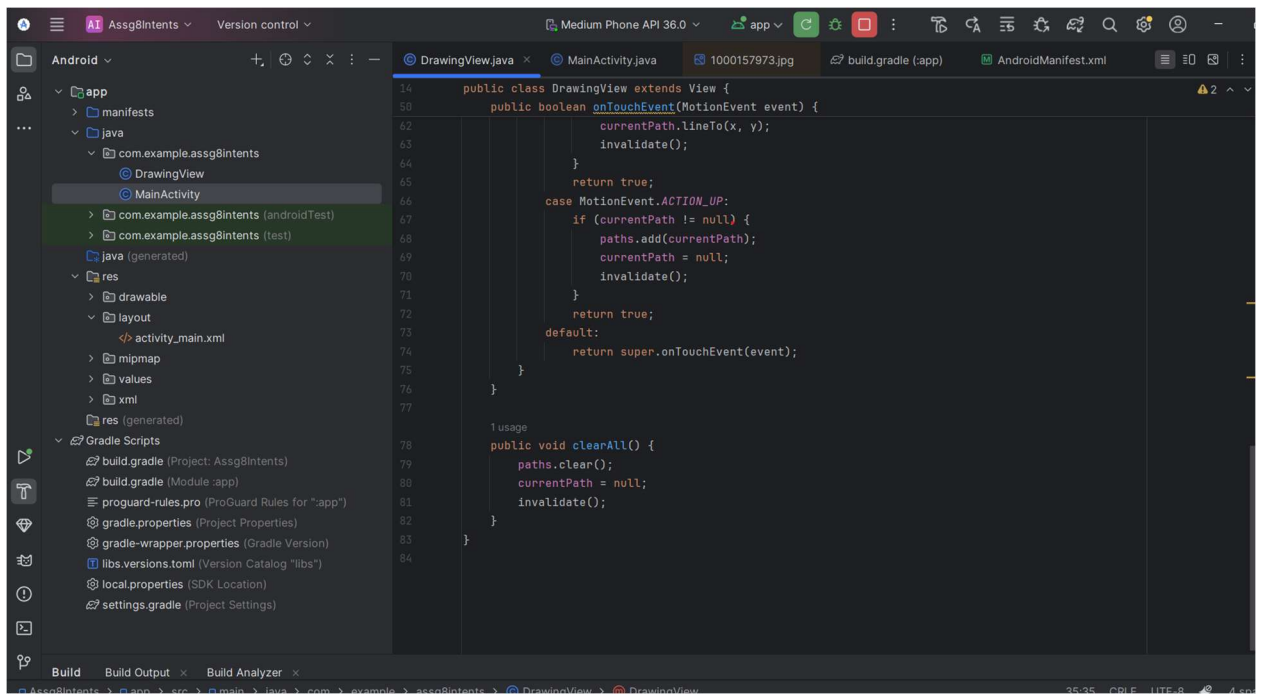
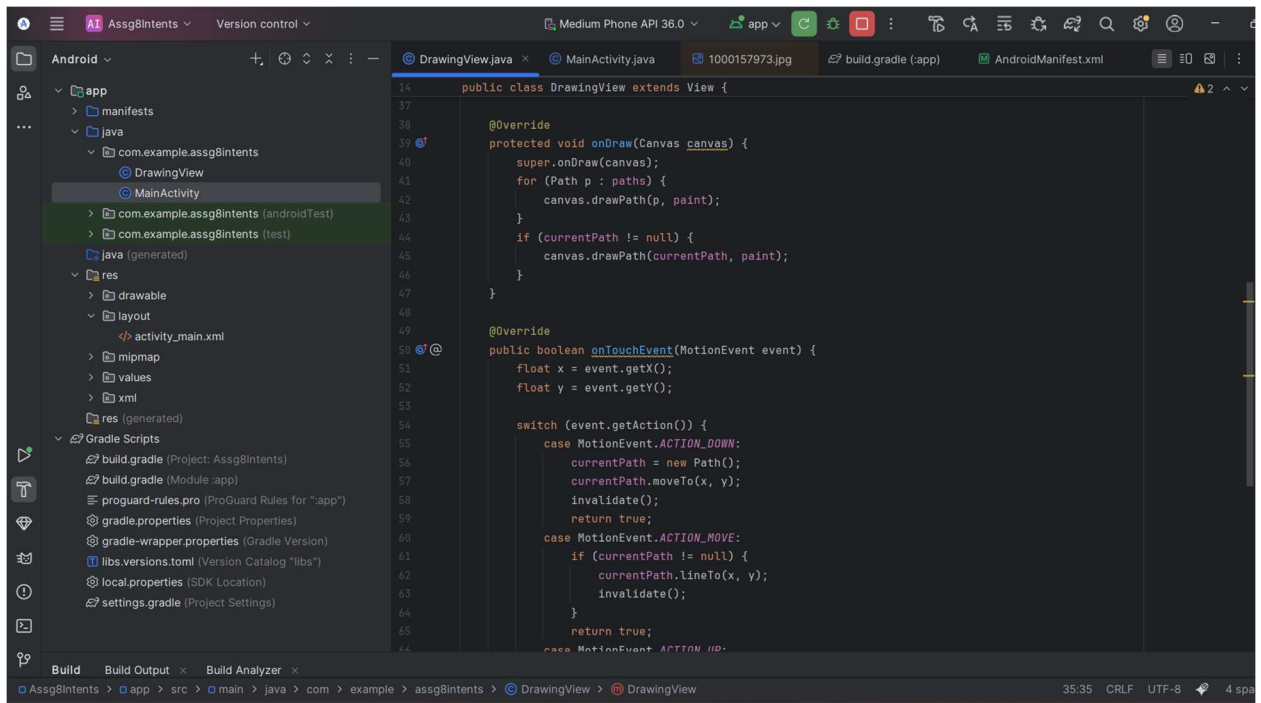


```
17 public class MainActivity extends AppCompatActivity implements ActivityResultCaller {
18     protected void onCreate(Bundle savedInstanceState) {
19         Button btnClear = findViewById(R.id.btnClear);
20
21         pickMedia = registerForActivityResult(
22             new ActivityResultContracts.PickVisualMedia(),
23             Uri uri -> {
24                 if (uri != null) {
25                     Log.d(TAG, "msg: PhotoPicker URI: " + uri);
26                     showImage(uri);
27                 } else {
28                     Log.d(TAG, "msg: No media selected");
29                 }
30             }
31         );
32
33         openDocument = registerForActivityResult(
34             new ActivityResultContracts.OpenDocument(),
35             Uri uri -> {
36                 if (uri != null) {
37                     Log.d(TAG, "msg: OpenDocument URI: " + uri);
38                     getContentResolver().takePersistableUriPermission(
39                         uri, Intent.FLAG_GRANT_READ_URI_PERMISSION
40                     );
41                     showImage(uri);
42                 } else {
43                     Log.d(TAG, "msg: No document selected");
44                 }
45             }
46         );
47     }
48 }
```

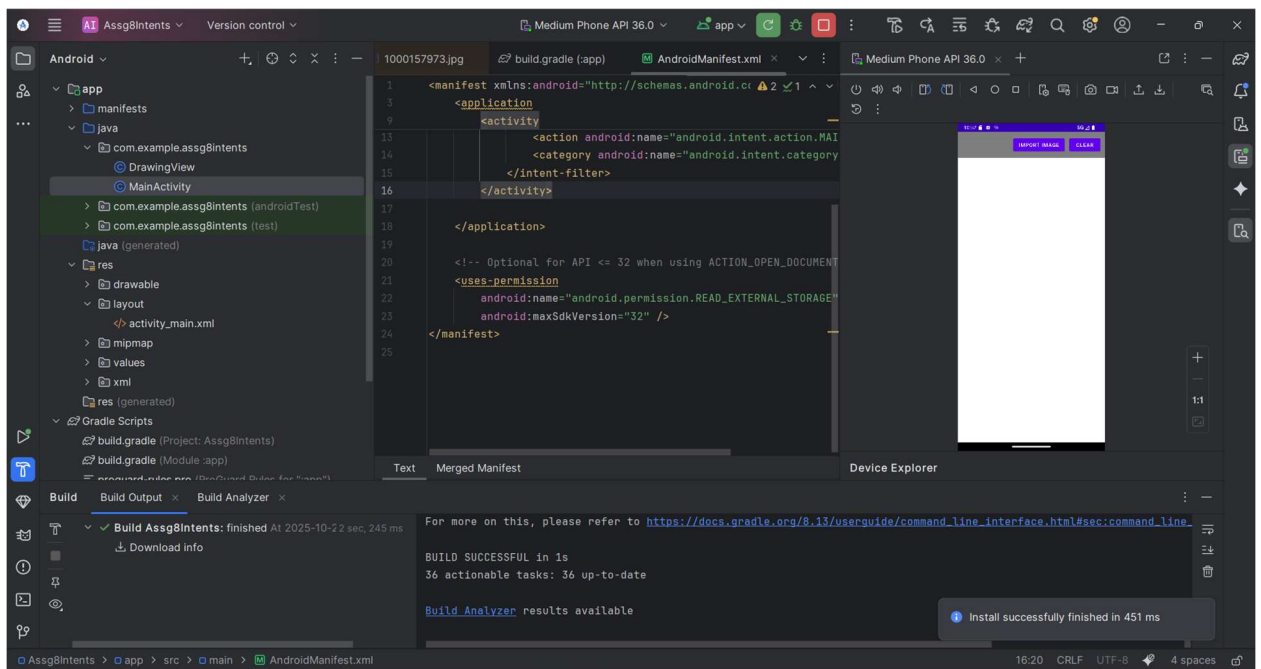
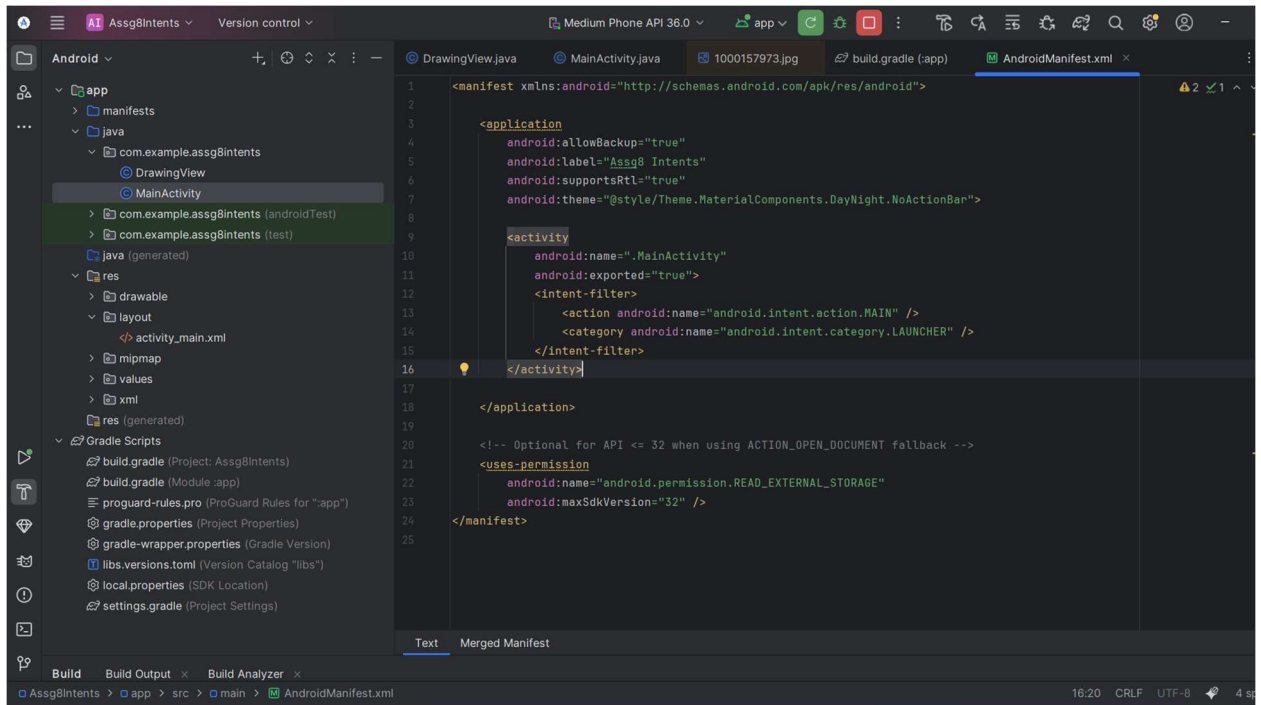


DrawingView.Java:

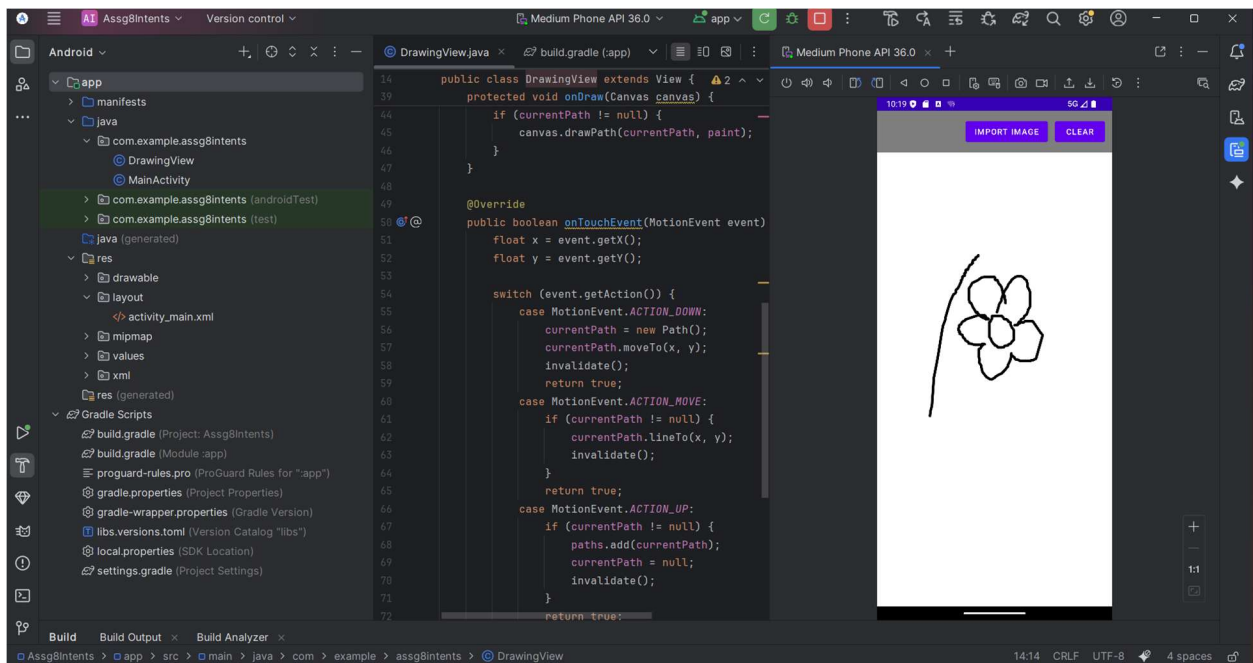
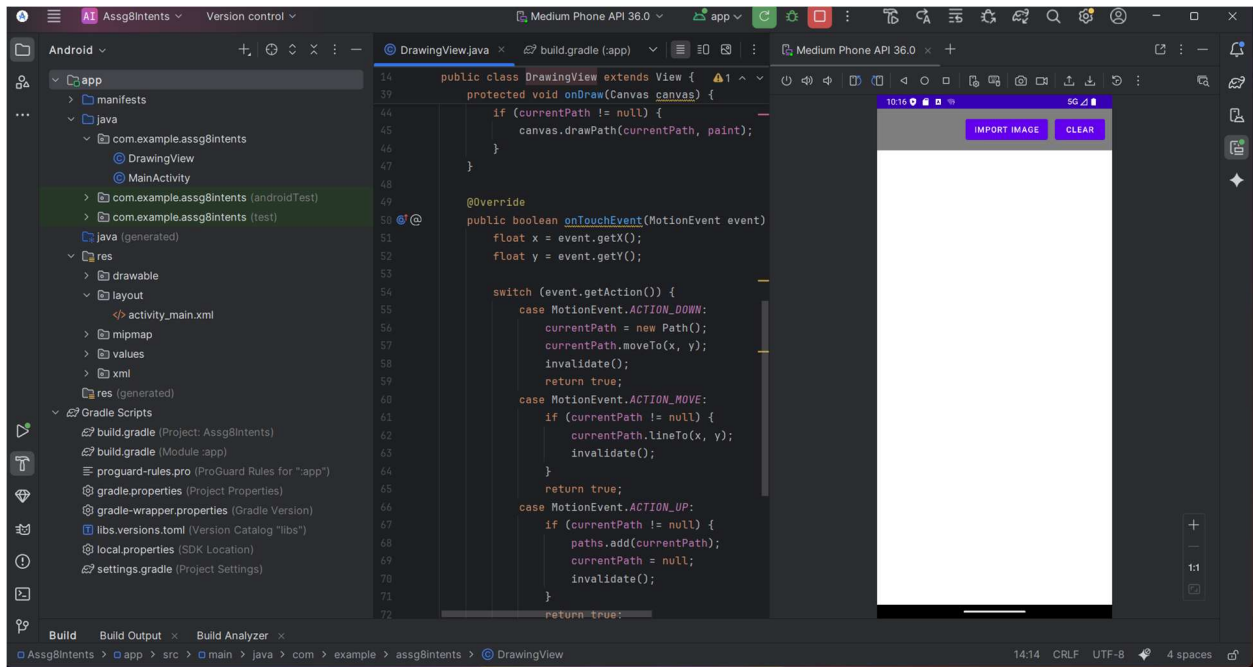




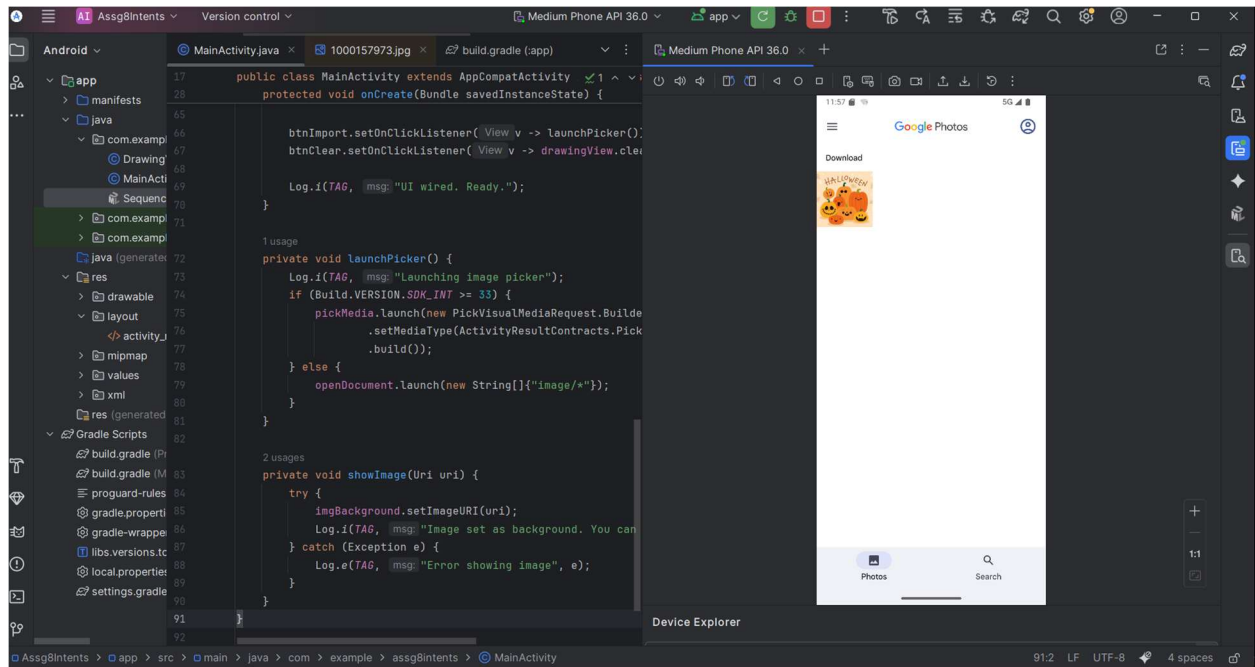
AndroidManifest.Xml:



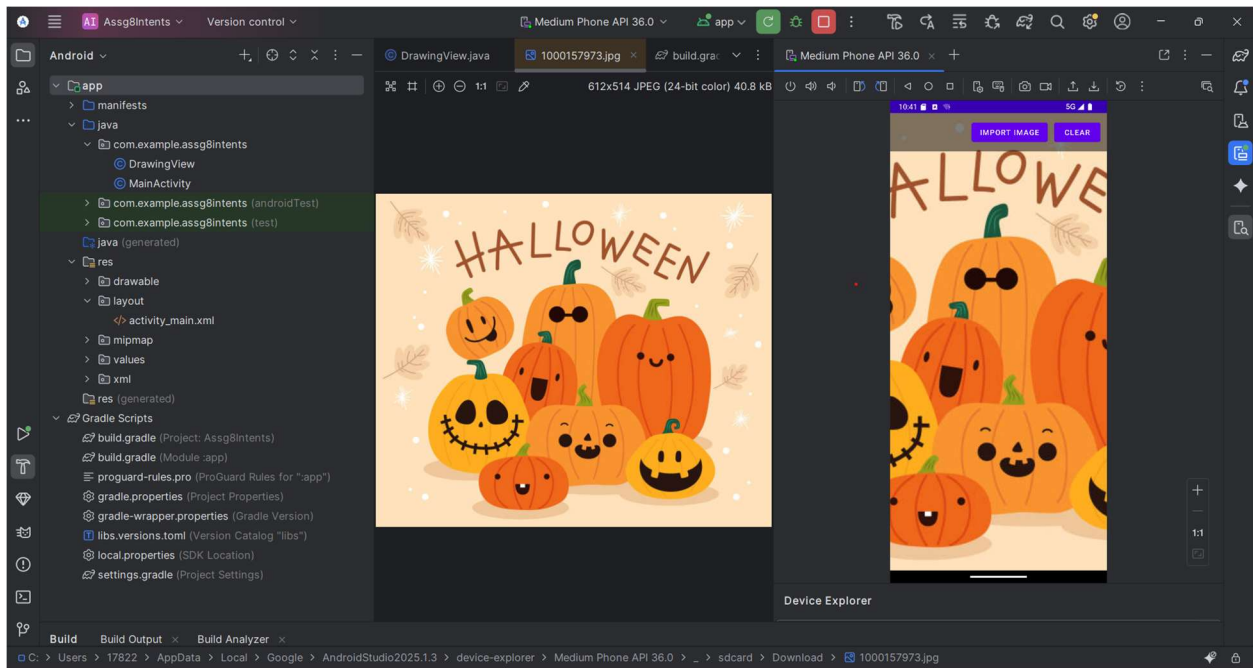
A blank screen can be drawn on at first (simple draw, just a line from a touch-drag, maybe a series of dots?)



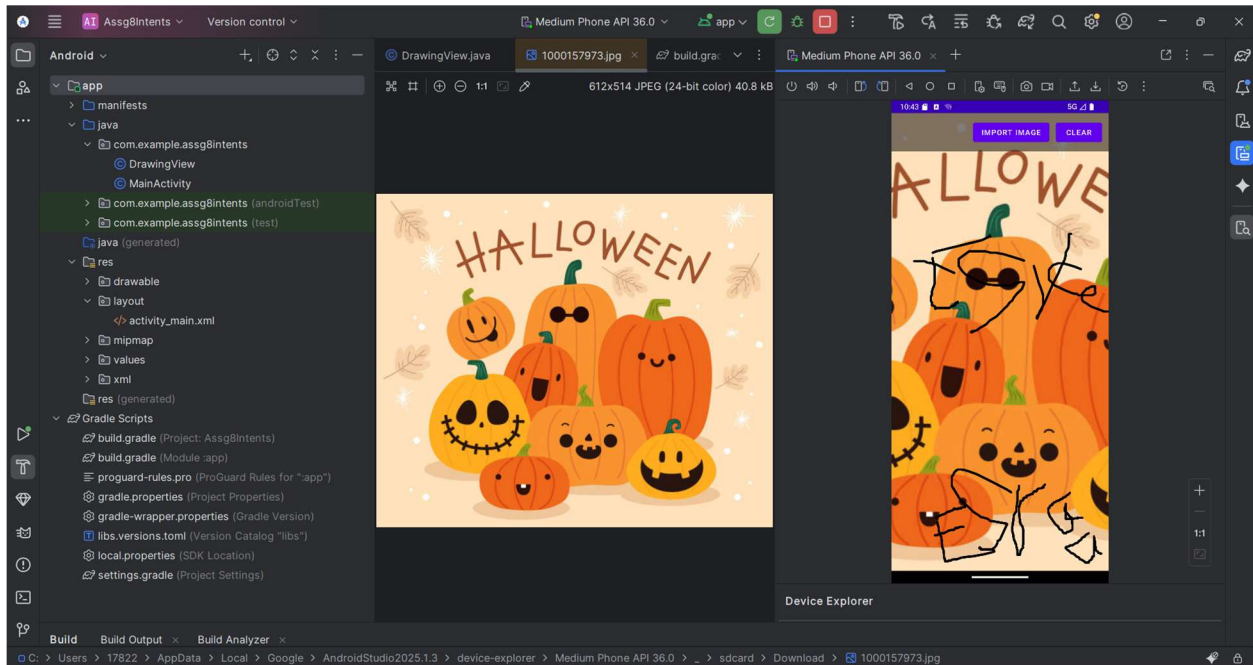
A picture can be imported via intents, from the photo gallery (you'll need to get some kind of picture in the gallery):



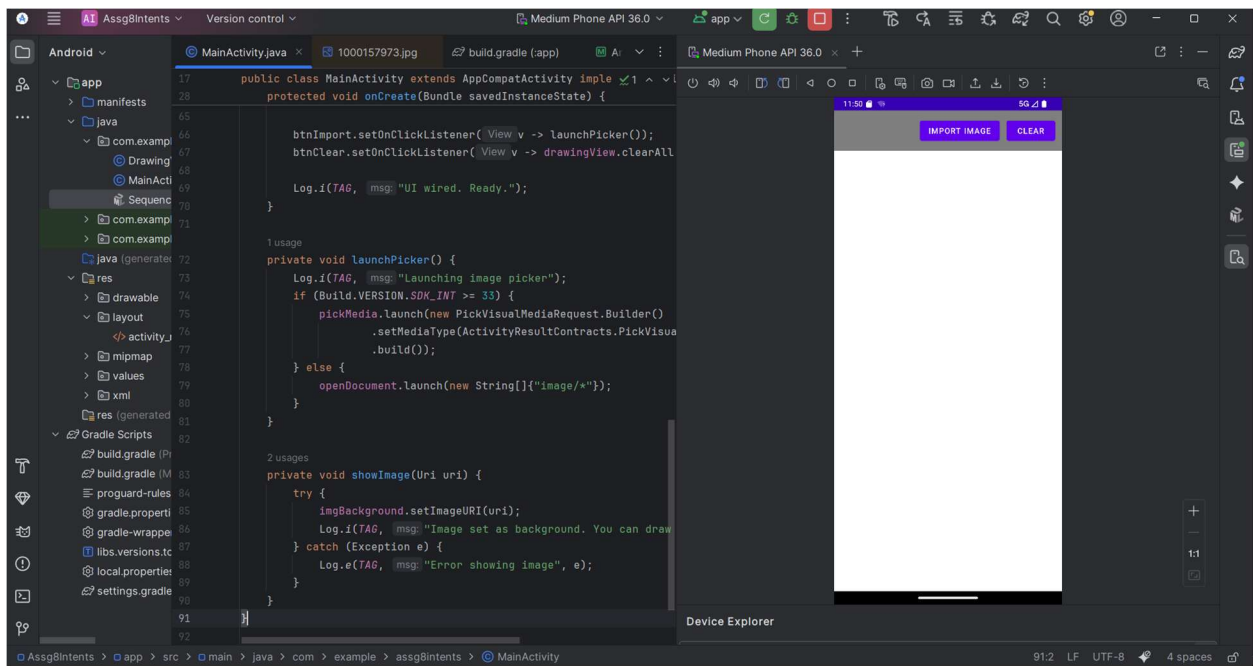
The picture is the background, and you can draw on top.



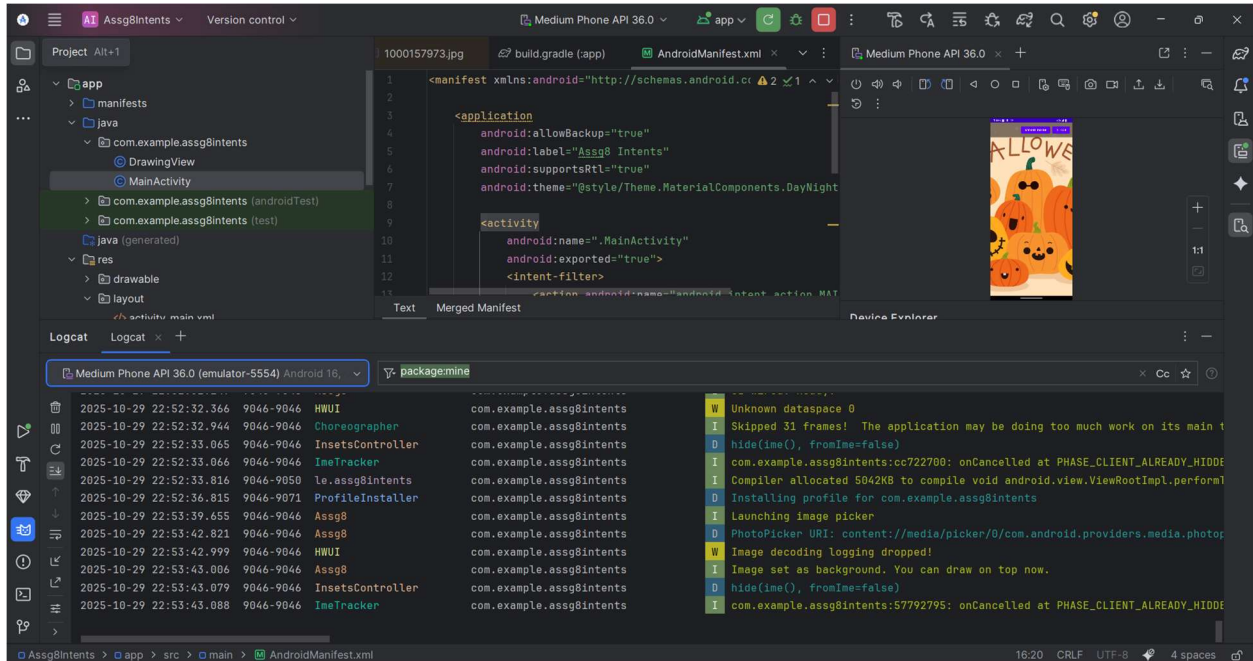
and you can draw on top:



The image need not be saved:

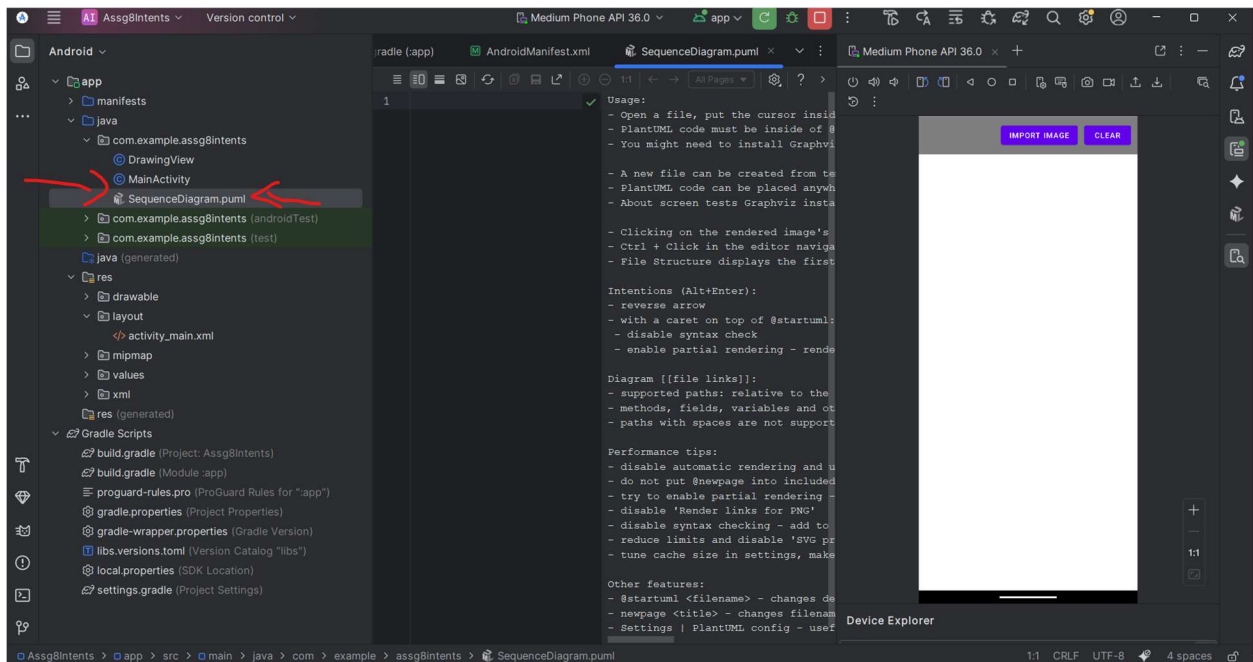
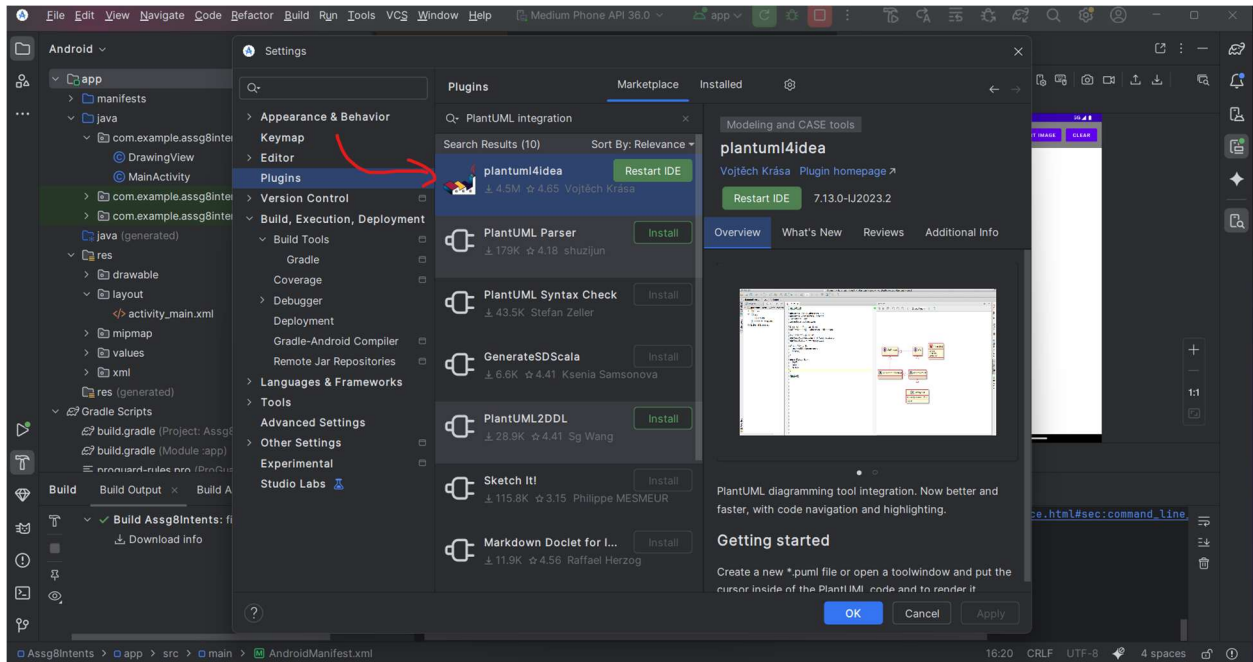


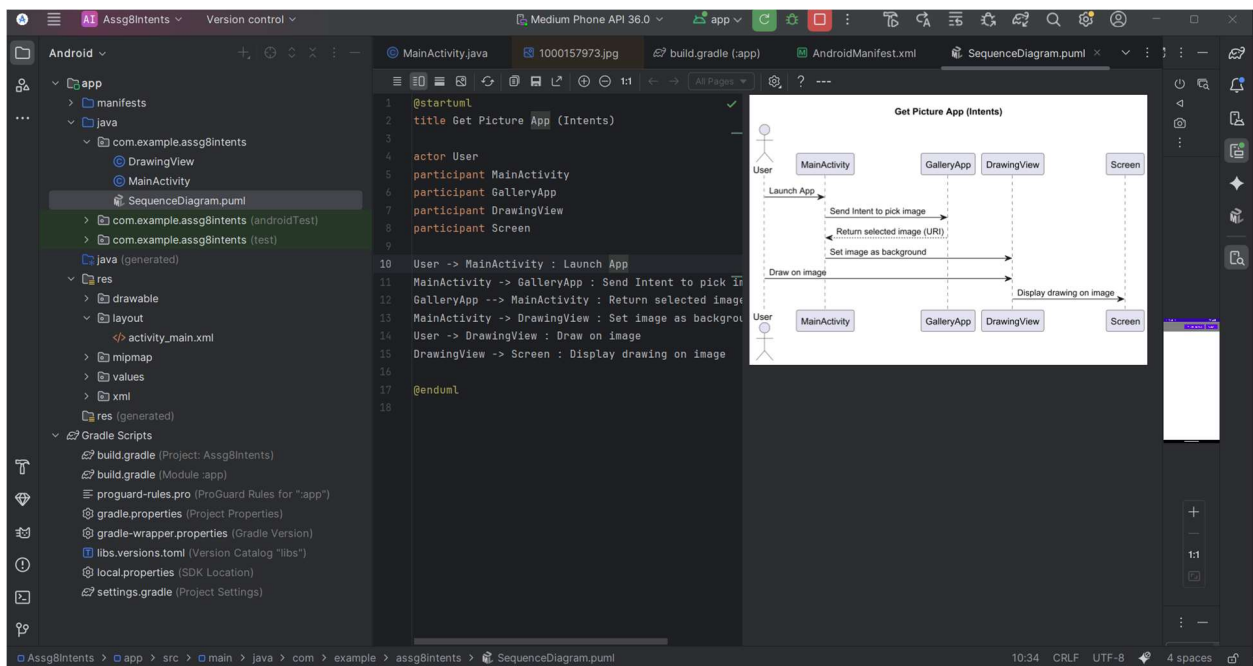
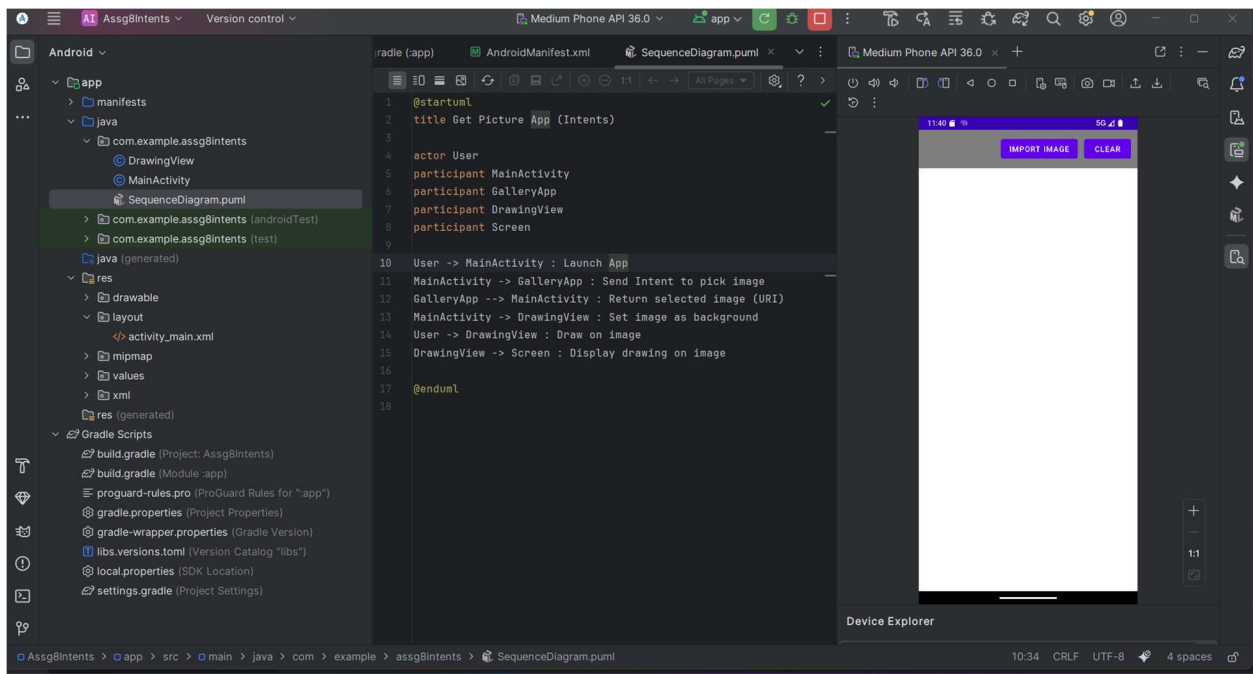
The LogCat:

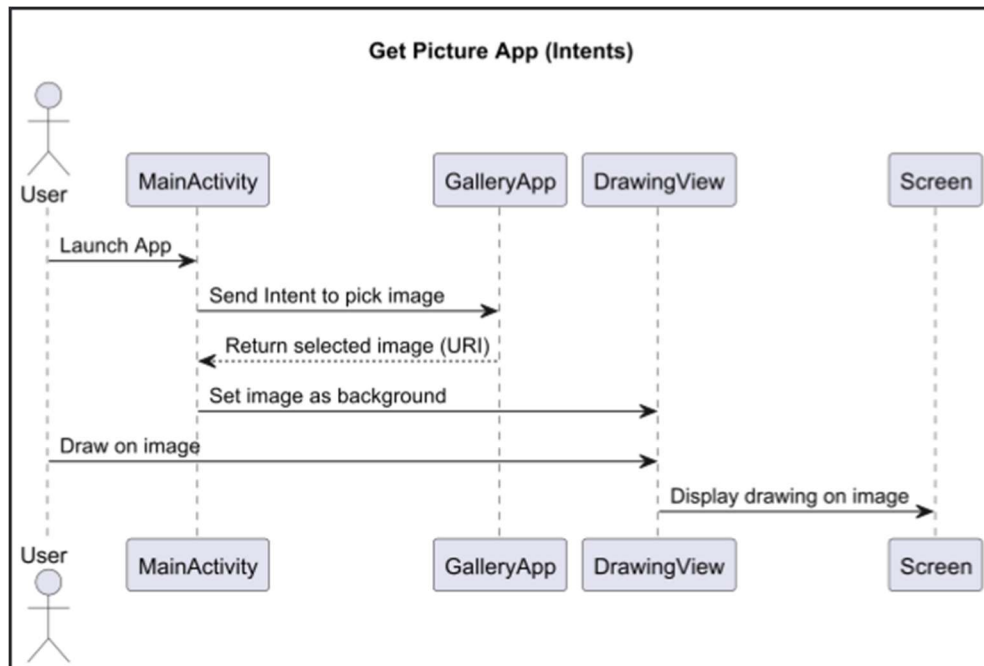


Logcat shows important messages from the app. These confirm that the Intent was launched and the image URI was received correctly

Create a suitable sequence diagram.







This sequence diagram explains the flow between the user, the MainActivity, the Gallery App, and the Drawing View when importing and drawing an image

Or from the site Hint2: [SequenceDiagram.org](https://sequencediagram.org) - UML Sequence Diagram Online Tool

